

XRPL Validator — Daily Engineering Report • 2026-04-10

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **6 commits** (xrpl-validator 6)
- 1 test / vector commit

Engineering changes

- **fix(xrpl-node): correct terPRE_SEQ sender extraction + shadow hash off-by-1** (dcc1d8e , xrpl-validator) Two diagnosis-driven fixes from a 63K-ledger battle test that exposed 557 FFI divergences and 3 shadow-hash mismatches: ffi_engine.rs — terPRE_SEQ recovery was extracting the wrong account. The scan looked for any 0x81 byte and read the next 20 bytes, but (a) random 0x81 bytes appear inside SigningPubKey/TxnSignature blobs which canonically come BEFORE sfAccount, and (b) the slice was off by one — it included the VL ...
- **feat(ui): cyberpunk PCB consensus board + sparkline transaction flow** (db09a15 , xrpl-validator) Two main visualization upgrades on the validator dashboard: Transaction Flow — replaced the static bar chart with a sparkline-style stacked area: 900 data points, 20ms update tick, smooth left-to-right fill with a glowing dot at the rightmost active sample. Top-4 tx types labeled inline. Persists to localStorage so the chart survives refreshes. Pattern lifted from the metrics.html sparkline and tuned for fluid ...
- **feat(xrpl-node): stage3 prep — config, batch helpers, hybrid singleton model** (9d9f694 , xrpl-validator) Adds the building blocks for Stage 3 (FFI bytes as the source of truth in state.rocks) WITHOUT wiring them into the live loop. The wiring is deliberately deferred until after Phase A — the 100K -> 200K -> 300K shadow-only battle test currently running on the validator host — clears with a clean shadow hash record. Flipping Stage 3 with even a small shadow mismatch rate would silently corrupt state.rocks for the divergent ledgers.
- **fix(bulk_sync): post-build root verification + seed-in-retention check** (0f8dcd7 , xrpl-validator) Investigation of the 91 missing SLEs from the prior battle test pointed at a class of silent-gap bugs in the bulk download path. None of the bugs were in our code per se — they're rippled-side partial responses that we trusted blindly: * ledger_data returned partial pages when the seed ledger had any leaves rippled hadn't fully cached locally * a worker hitting >50 errors silently gave up mid-walk * the SHAMap ...
- **refactor(xrpl-node): centralize data paths in xrpl_node::paths** (2d8ecca , xrpl-validator) The data-directory layout was split-brain across the codebase. bulk_sync read XRPL_DATA_DIR (with a buggy default that included /sync), while live_viewer hardcoded /mnt/xrpl-data/... in nine places. The split was masked on the validator host by a /mnt/xrpl-data -> ~/xrpl-data symlink, so nothing visibly broke — but a fresh deployment without the symlink would have had the writer and the reader pointing at ...

Tests and campaigns

- **test(xrpl-node): pin terPRE_SEQ extraction + shadow overlay against regressions (4f384a5 , xrpl-validator)**

Appendix — commits

Time	Repo	Commit	Subject
19:26	xrpl-validator	dcc1d8e	fix(xrpl-node): correct terPRE_SEQ sender extraction + shadow hash off-by-1
19:27	xrpl-validator	db09a15	feat(ui): cyberpunk PCB consensus board + sparkline transaction flow
19:45	xrpl-validator	4f384a5	test(xrpl-node): pin terPRE_SEQ extraction + shadow overlay against regressions
19:46	xrpl-validator	9d9f694	feat(xrpl-node): stage3 prep — config, batch helpers, hybrid singleton model
19:56	xrpl-validator	0f8dcd7	fix(bulk_sync): post-build root verification + seed-in-retention check
20:06	xrpl-validator	2d8ecca	refactor(xrpl-node): centralize data paths in xrpl_node::paths